

Module 7A - Our Cosmic Neighborhood

Module Structure

This module is a **viewing and learning module**, meaning we will be able (or try) to view certain celestial objects in the night sky.

Two notable events occurring soon:

- the Full Moon ([Harvest Moon](#)) occurring on September 26th, and
- the Autumnal (or [Fall Equinox](#)) occurring on September 22nd.

Clear skies permitting, we will try to view these celestial bodies over the next week:

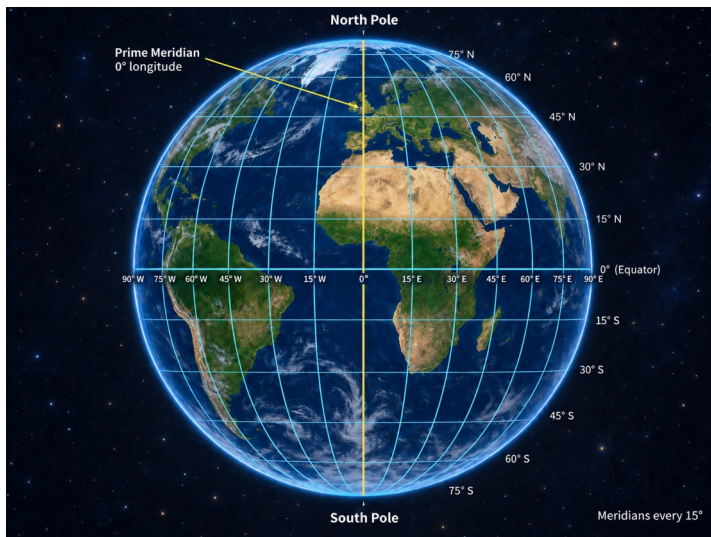
- the Full Moon
- the Orion asterism, and Orion's Belt and Scabbard
- the Summer Triangle (the stars Altair, Deneb and Vega), and
- the planets Venus and Mars

Preliminary Considerations

Before getting into actual viewing, it is worth discussing two important ideas in astronomy. The **first important idea** is **how clock time works** worldwide.

In the United States, we use the 12 hour clock time convention, am and pm. Elsewhere, the 24 hour clock time convention is very often used professionally, with the 12 hour clock time convention (am, pm) often used casually.

The 24 hour clock is exclusively used in astronomy for Earth based time calculations.



Because Chicago's longitude is almost 90 degrees west of the [prime meridian](#), or 0 degrees longitude, the longitude going through London, English clock time in Chicago is delayed by a quarter of a day, or 6 hours. Chicago is typically 6 hours behind London.

The British or London 24 hour clock is formally known both as Greenwich Mean Time, GMT, and Universal Time Coordinated, UTC.

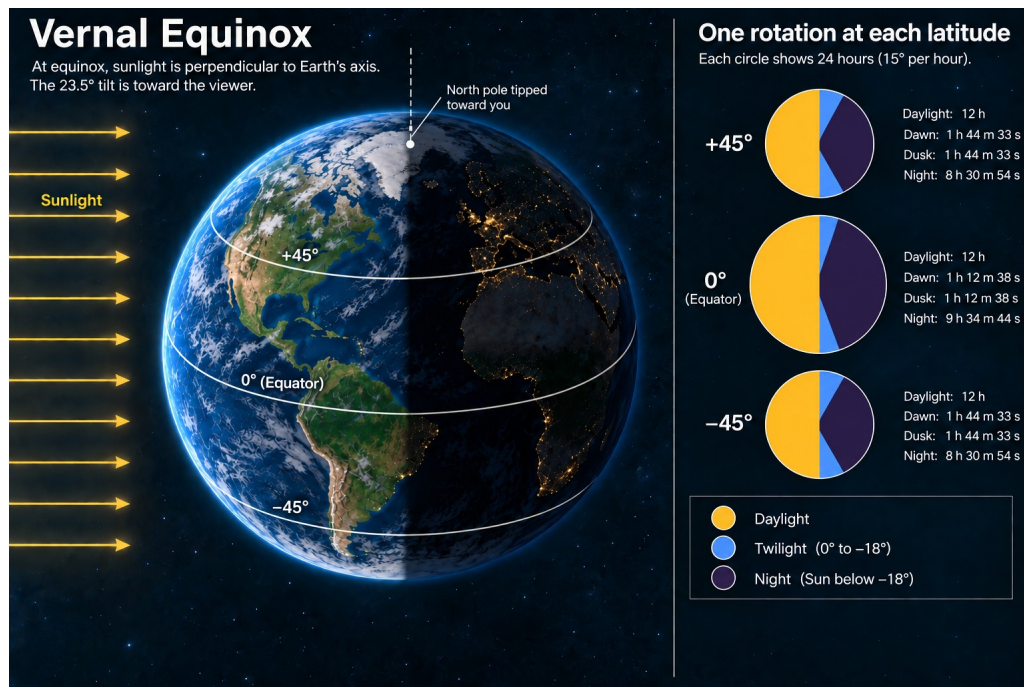
Right now, here in Chicago, **until 11/1/26**, we enjoy Central Daylight Saving Time (CDT), so our local time is an hour ahead of our normal Central Standard Time (CST).

Right now, in the UK, **until 10/25/26**, Londoners enjoy their equivalent of daylight savings time, known as British Summer Time (BST).

For example, right now if it were 12:00 am BST in London (midnight, the start of a new day there), it would be 6 pm CDT in Chicago, the previous day.

The **second important idea** is the **Autumnal Equinox Moment**. This is the moment, London time UTC, when the Sun, on its fictitious annual journey around Earth, crosses over the Equator, from the Southern Hemisphere to the Northern Hemisphere. The Autumnal (or Fall) Equinox Moment this year, for England, is 00:06 GMT, September 23, 2026; simultaneously, in Chicago, that's 7:06 pm CDT, September 22, 2026.

The **Autumnal Equinox Day** is the day the **Autumnal Equinox Moment** happens locally. It is the **official first day of fall, astronomically speaking**.



On either of the two Equinox Days (there's two, one for fall and one for spring), the Sun shines more or less directly over the Equator. If you were living on the Equator, at noon you would see the Sun shining directly over your head. (We say it would be shining **zenially**.)

[Here we are looking at the Vernal Equinox, but the following argument is the same for both equinox days.]

Because on an equinox day the Sun is shining directly over the Equator, its shadow line of illumination of the Earth passes directly through Earth's North and South Poles. It follows that **every person on Earth** experiences (roughly) 12 hours of direct sunshine, and 12 hours of twilight plus nighttime.

From the [time and date website](#), we see that the time interval between sunrise and sunset in Chicago on this year's Fall Equinox Day is a bit longer than exactly 12 hours (by 9 minutes).

2026	Sunrise/Sunset		Daylength		Astronomical Twilight		Nautical Twilight		Civil Twilight		Solar Noon	
Sep	Sunrise	Sunset	Length	Diff.	Start	End	Start	End	Start	End	Time	Mil. mi
20	6:35 am → (88°)	6:51 pm ← (272°)	12:15:02	-2:47	5:01 am	8:24 pm	5:35 am	7:51 pm	6:06 am	7:16 pm	12:43 pm (49.0°)	93.346
21	6:37 am → (88°)	6:49 pm ← (271°)	12:13:15	-2:47	5:03 am	8:22 pm	5:36 am	7:49 pm	6:09 am	7:17 pm	12:43 pm (48.6°)	93.322
22	6:38 am → (89°)	6:47 pm ← (271°)	12:09:28	-2:47	5:04 am	8:21 pm	5:37 am	7:47 pm	6:10 am	7:15 pm	12:43 pm (48.2°)	93.295

23	6:39 am → (90°)	6:45 pm ← (270°)	12:06:41	-2:47	5:05 am	8:19 pm	5:38 am	7:46 pm	6:11 am	7:13 pm	12:42 pm (47.8°)	93.269
24	6:40 am → (90°)	6:44 pm ← (270°)	12:03:53	-2:47	5:06 am	8:17 pm	5:39 am	7:44 pm	6:12 am	7:11 pm	12:42 pm (47.5°)	93.243
25	6:41 am → (91°)	6:42 pm ← (269°)	12:01:06	-2:47	5:07 am	8:15 pm	5:40 am	7:42 pm	6:13 am	7:10 pm	12:42 pm (47.1°)	93.216
26	6:42 am → (91°)	6:40 pm ← (269°)	11:58:19	-2:47	5:08 am	8:13 pm	5:41 am	7:40 pm	6:14 am	7:08 pm	12:41 pm (46.7°)	93.190

Rise
6:35 am
88°E

Meridian
12:43 pm
180°S

Set
6:51 pm
272°W

12:43 pm

Altitude
49°

Heading
↓ 180° S

Position
Day

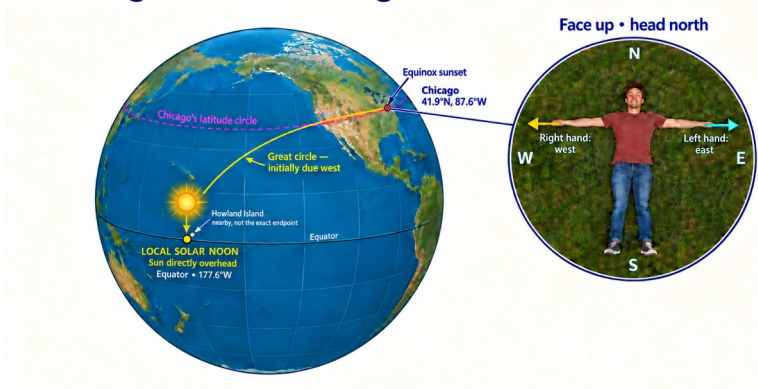
This discrepancy is due to the fact that the Earth is not a perfect sphere.

Of course, the sky does not immediately become bright when the Sun rises over the easterly horizon (or become dark when the Sun sets over the westerly horizon). The sky and ground are illuminated indirectly by the Sun, with sunlight bouncing off the atmosphere for a moderately short period of time before the Sun rises (**dawn**), and for an identical short period of time after the Sun sets (**dusk**).

These indirectly lighting periods are collectively called **Twilight**.

For Chicago, on both equinox days (Spring and Fall), the twilight in the morning, dawn, and the twilight in the evening, dusk, are both about a half hour each. It follows that the **total amount of lighting**, directly and indirectly, from the Sun in Chicago on either equinox day is about 13 hours.

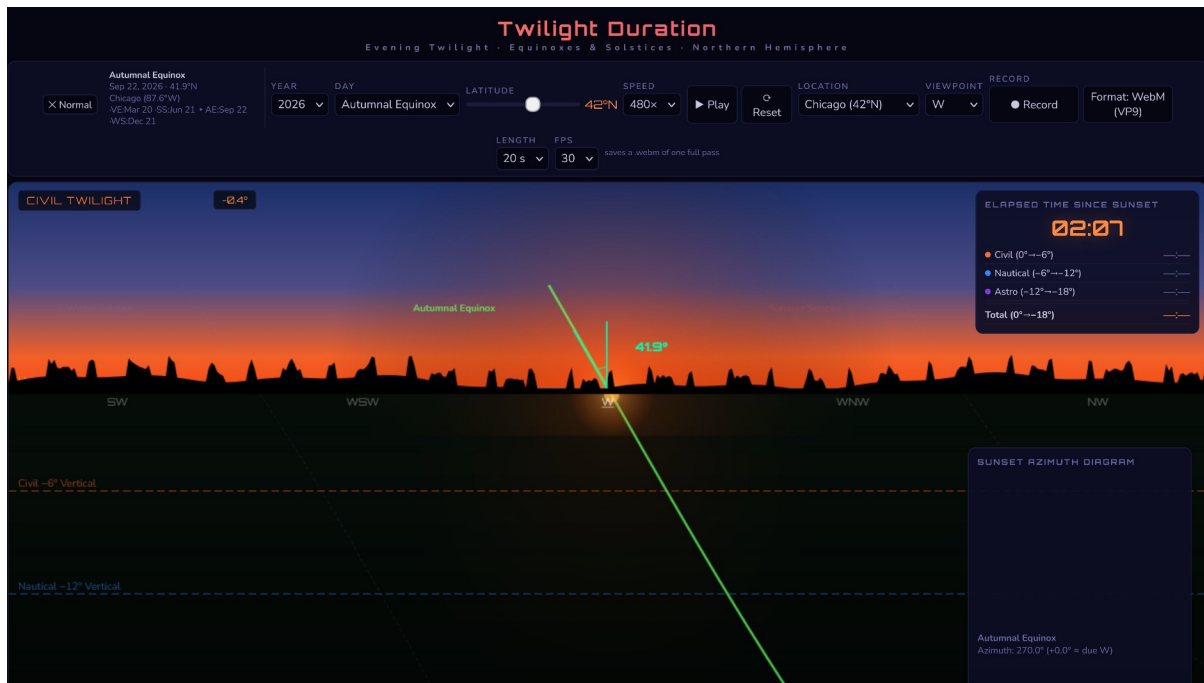
Pointing west from Chicago



Another equinox day observation worth mentioning is that the Sun sets directly over cardinal west for **everyone on Earth**.

At first this might seem impossible, until you realize that when you, in Chicago, on this day, point west at the setting Sun, you are actually pointing along a [great circle arc](#) to a point longitude 180° west on the Equator. Basically, you are pointing at the intersection of the Equator and the [International Date Line](#) (about Howland Island, where [Amelia Earhart's](#) plane was lost), the point where the Sun is directly overhead (local noon).

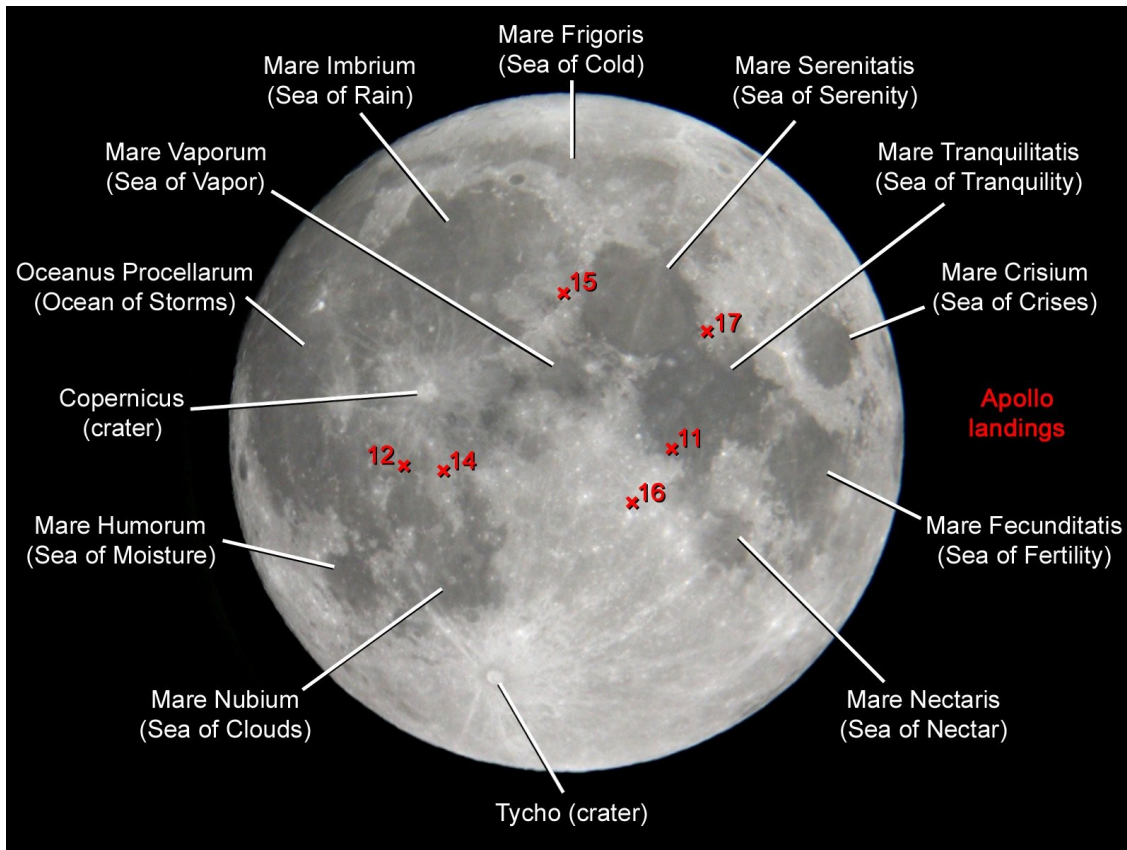
One final observation worth mentioning about the equinox days is that you can use the setting Sun to **determine your latitude on Earth**.



Since the Sun is setting due west, the celestial path of the Sun can easily be determined. This path, and the vertical line up from the point of the Sun's impingement on the westerly horizon make an angle **that is your latitude**.

Looking at the Moon

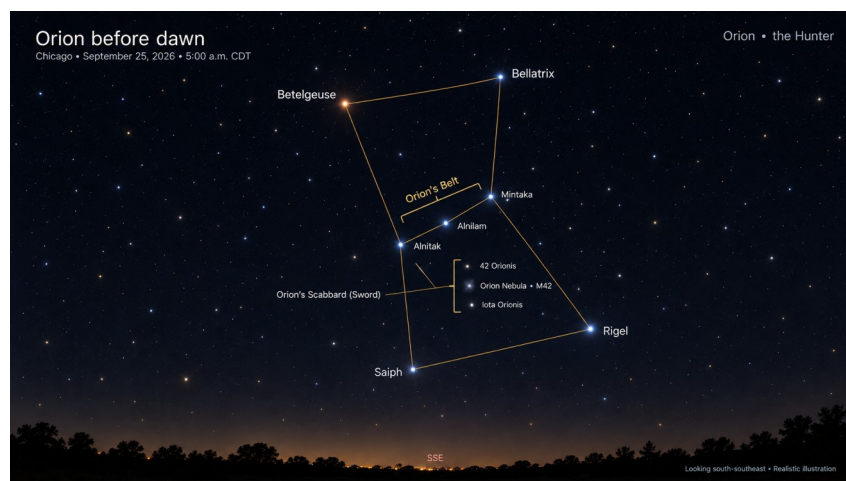
With the unaided eye, or by telescope or binoculars (if you have either), look for the Full Harvest Moon on the evening of September 26, 2026. Below we see the Full Moon, in all its glory, with the six Apollo lunar missions' landing locations shown in red. Note that there is no landing location for Apollo 13. Do you know why?



Here are some questions to ponder on, while observing the Full Moon:

- Does the Moon look gauzy when close to the horizon? Is so, why?
- If viewing with a telescope or binoculars, does the image you see seem upside down, relative to the picture above? Are right and left switched relative to the picture above?
- Where is Tycho? Where is Copernicus?
- Where did Apollo 11 land?

Orion and His Belt and Scabbard

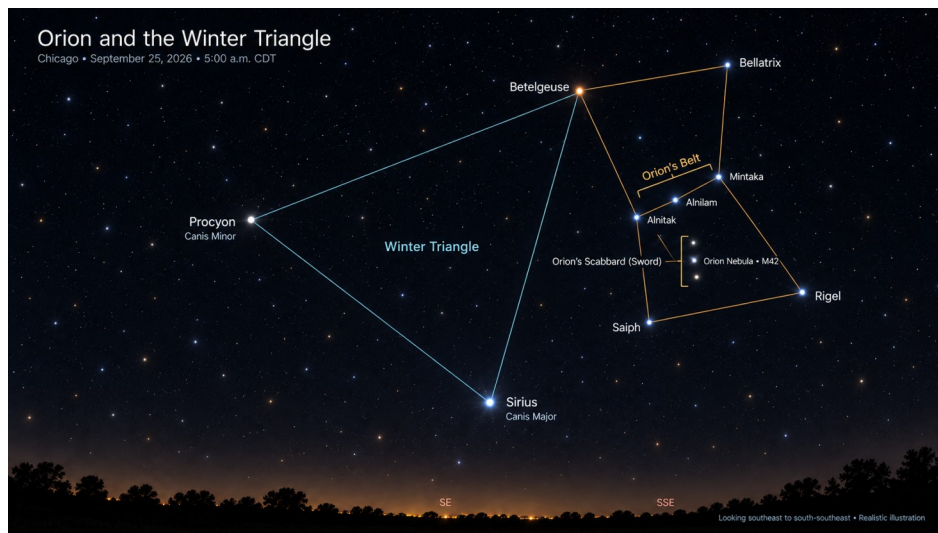


Orion is both a [constellation](#) (patch of sky) and an [asterism](#) (star pattern). You should be able to see the asterism Orion (a warrior) in the SSE sky in the early morning of 9/25/26. Within the Orion asterism, look for three horizontal stars that look like a belt. [These three stars are called [Orion's Belt](#), but are also known as “The Three Kings” and “The Three Sisters.”] Hanging below the belt vertically are an additional three stars called [Orion's Scabbard or Sword](#). [These three stars are very hard to see, but your smartphone can capture them with ease.]

Visible from most everywhere on Earth, the constellation Orion (Orion the Hunter) looks like a male warrior facing away from you, the observer.

The Winter Triangle

While you're looking at Orion and his belt and sword, also look for the **Winter Triangle**, an asterism composed of three stars that form an equilateral triangle.



The three stars are **Betelgeuse, Sirius and Procyon**.

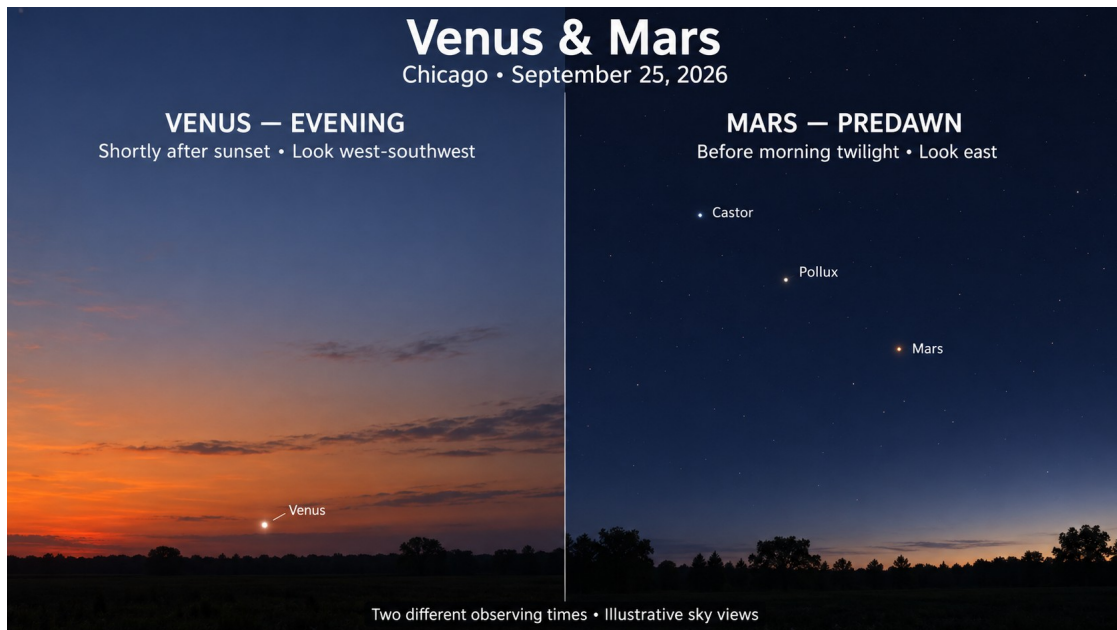
[Betelgeuse](#) (pronounced beetle juice) is Orion the Hunter's left shoulder. Recall that Orion, the Hunter, is said to be facing away from you.

Betelgeuse is about 640 light years (ly) away. It is only about 10 million years old, but because of its original (blue star) size, it is **already dying** (and is now a [Red Supergiant](#) star). It is expected to explode as a [supernova](#) in a few million years.

[Sirius](#), commonly known as “The Dog Star,” is actually a binary, or two star system. It is in the constellation [Canis Major](#) (“Greater Dog”), and is quite close to us, only about 8.6 light years (ly) distance. It is the **brightest star in the night sky**.

[Procyon](#), in the constellation [Canis Minor](#) (“Lesser Dog”), is the **eighth brightest star** in the northern hemisphere's night sky. It is also a binary star system, and is also quite close to us, at 11.5 ly distance.

The Planets Venus and Mars



At this time, we see the planet Venus in the westerly evening sky, in the civil twilight period (9/25). By contrast, we see the planet Mars in the following early morning (night) sky (9/26).

The Story of Venus

[Venus](#) is the second planet from the Sun, a terrestrial or rocky planet, and is the **second brightest** reflecting object in the solar system, after the Moon.

Venus is about 0.7 AU from the Sun. [One AU is the average Sun-Earth distance, about 93 million mile.]

The Greeks called this version of Venus the evening star [Hesperus](#), with the Romans calling this version of Venus the evening star Vesper.

Since Venus is an inferior planet (a planet closer to the Sun than Earth), it can also appear in the morning, making it a morning star. The Greeks called Venus the morning star Phosphorus, with the Romans calling it [Lucifer](#).

A serious student of astronomy soon learns that Venus is *not* for lovers, but is instead a miserable place, with an atmosphere that is 96% carbon dioxide and 4% nitrogen. Venus' atmospheric pressure is a crushing 93 bar, or 92 Earth Atmospheres. (One bar is 100,000 Pascals.)

Venus' surface temperature is a steady 872 degrees Fahrenheit, day and night, due to a runaway greenhouse gas effect. This temperature is high enough to melt lead, and so desolder all the regular electronics in your space ship.

Venusian clouds are made of sulfur dioxide and microdroplets of sulfuric acid. Lightning in those clouds accompanies your descent to Venus' waterless surface.

Venus has no magnetic field to protect you from the solar wind particles striking your cells' DNA, but there have been reports of auroral activity on Venus induced by the solar wind.

Air speed at the top of the Venusian cloud cover is a staggering 220 miles per hour, but at the surface of Venus it is a mere 2.5 miles per hour.

A Venusian day can be pretty gloomy too: only 5,000 to 14,000 lux of sunshine illuminates Venus' surface, compared to 100,000 lux of sunshine on a bright Earth day.

Your gloomy Venusian day, accompanied by the occasional volcano, will be long, roughly 117 Earth days. With a Venusian year being about 225 Earth days, each Venusian year is composed of only about two Venusian days, because the planet **rotates clockwise**.

One good thing though for you, our planetary messenger: as you struggle to move about on Venus, **you will weigh about 16% less**, 10% less from the g of Venus (8.8 meters per second squared vs 9.8 meters per second squared here on Earth), and 6% less from the buoyant force of the atmosphere itself.

[The buoyant force on you from the atmosphere here on Earth is, by comparison, a mere one eighth of one percent of your weight, or less than one Newton out of 681.]

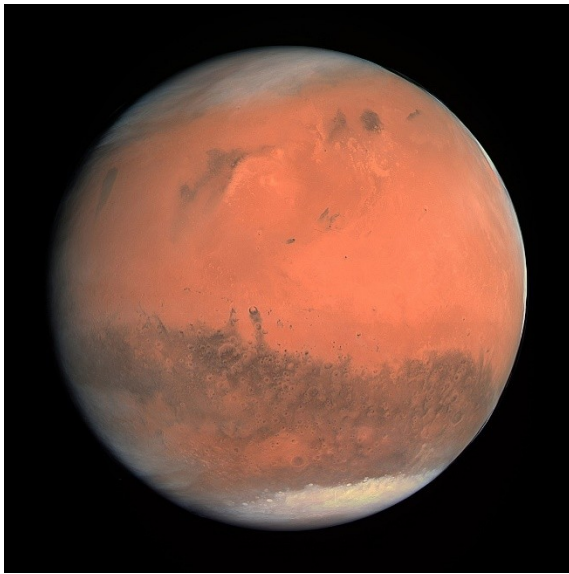
Note: Soviet spacecraft (Venera missions) have landed “successfully” on Venus on a few occasions.

The Story of Mars

Mars is often referred to as the **red planet**, because iron oxide covers its surface. It is the **fourth brightest sunlight reflecting object** in the Solar System, surpassed only by the Moon, Venus and Jupiter.

Mars has two “asteroid” moons, Deimos and Phobos.

Mars has no active magnetosphere, and so its surface is subject to solar wind bombardment. There are “magnetic umbrellas” near Mar's south pole, and aurorae occur planetwide.



While there may be some residual volcanic activity on Mars, the most prominent volcano on Mars is the dormant [Olympus Mons](#). At 72,000 feet tall, it is two and a half times taller than Mount Everest on Earth. It is the **largest planetary volcano** in the Solar System.

Mars has an axial tilt and rotation period quite similar to Earth. A Martian day is a little less than **25 Earth hours**, and Mars exhibits four seasons. Since Mars is about 1.5 times farther away from the Sun than the Earth, its year last a little less than **2 Earth years**, and so its seasons are about twice as long as Earth's.

The atmosphere of Mars is, on average, quite thin, measuring **less than one percent** of the Earth's atmosphere at sea level. As such, very little solar warmth is retained during the Martian day or night, with daily surface temperatures ranging from **-225° F to 95° F**, depending on location, season, and day-night period.

First Extended Study: [The Moon](#) is almost as old as the Earth, with an age of about 4.51 billion years. (The Earth is 4.543 billion years old.)



Most astronomers concur that the **Moon was created by a collision** between a [planetismal](#) (a small planet-like object called **Theia**) and Earth. It is thought that this planetismal struck the Earth with a glancing blow.

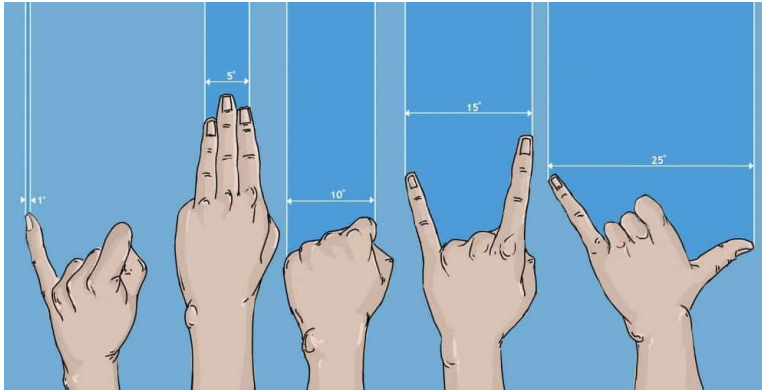
The planetismal was vaporized in its collision with Earth (along with a chunk of Earth). Remnants of the collision then recombined into a **ring-like form** girdling the Earth. That ring form then gravitationally and electrostatically consolidated into a spherical object circling the Earth, thus becoming the Moon we know.

The Moon was quite close to the Earth in its infancy, but over a long period of time, it “stole” rotational kinetic energy from Earth (through tidal action). This “theft” of energy caused the Earth's rotation to slow, and the Moon to “migrate out” to orbits farther and farther away from Earth.

The Earth-Moon orbital radius at Moon “birth” was probably about 15,000 miles center to center.

By contrast, the **current Earth-Moon distance**, is approximately 239,000 miles (on average, center to center), about 16 times that first orbital radius. This distance is

regularly measured by shining a laser beam off one of multiple reflecting mirrors placed on the Moon by the Apollo astronauts and Russian rovers.



While we know the diameter of the Moon from having visited it numerous times, observers on Earth sometimes prefer to describe the size of most any celestial body, including the Moon, by what **angle it makes** in the sky compared to an observer's body part.

It turns out that your **pinky finger**, at arm's length, subtends an angle of about 1° .

The Moon surprisingly subtends only half that angular size. So even though the Moon looks huge (because of the [Moon Illusion](#)), your pinky finger, at arm's length, easily covers over the Moon completely.

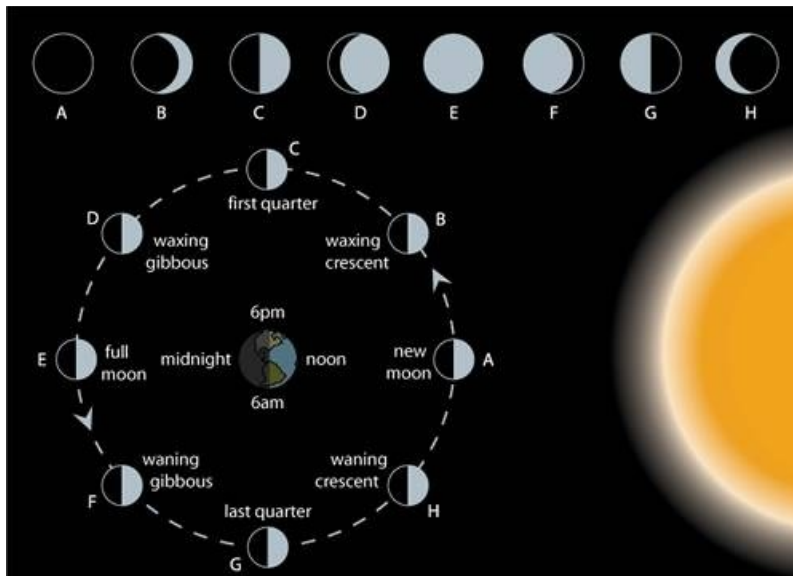
Because the Moon and the Earth have mutually experienced tidal forces between them for billions of years, the Moon is **tidally locked** with the Earth, meaning that the Moon shows only one side of itself to an Earthbound observer. That side is called **the near side** of the Moon.

The other side of the Moon is called **the far side**. Contrary to what many people have been led to believe, the far side of the Moon is not dark at all: we just don't see it from Earth.

Because of tidal locking, the **Sidereal Moon Day** (the time for the Moon to rotate once on its axis, compared to the faraway stars) is the same as the **Sidereal Month** (the time for the Moon to orbit the Earth, compared to the faraway stars). The Sidereal Moon Day and Month are **27.25 days**.

For an observer on Earth, the idea of a Month however is firmly connected with a **lunation**, or **Synodic Month**, the time between a New Moon (no sunlit illumination of the Moon) and the next New Moon.

Since the Earth moves through its orbit during a lunation, the Earth must rotate a bit more than two more times for an Earthbound observer to see a New Moon again. It follows that a lunation period, or Synodic Month, is longer than a Sidereal Month, and is in fact **29.5 days**.



During a lunation, we see the near side of the Moon in various states of solar illumination, which we call **phases**.

In the image to the left, we see the Moon in **eight phases** of illumination, as viewed by an Earthbound observer looking at the sky at 6 pm, midnight or 6 am, standard time.

Some conclusions (applicable to the Northern Hemisphere):

At 6 pm local standard time, the New Moon is setting over the westerly horizon.

At 6 pm local standard time, the First Quarter Moon (in the Northern Hemisphere) is seen due South.

At 6 pm local standard time, the Full Moon is seen rising due East.

At midnight local standard time, the Third Quarter (or Last Quarter) Moon is seen rising due East.

At 6 am local standard time, the Third Quarter (or Last Quarter) Moon is seen due South (in the Northern Hemisphere, whether the Sun is out or not).

[Note: The only way you could see the Third Quarter Moon at 6 pm local standard time is if you were standing at the North or South Pole.]

The New Moon, First Quarter Moon, Full Moon and Third Quarter Moon are considered the **main phases of illumination** of the Moon.

The four other (minor) Moon phases (in order from New Moon to New Moon) are:

waxing crescent, waxing gibbous, waning gibbous, and waning crescent.

Study the chart above to master the full sequence of lunation.



It's true you can tell what season it is from the calendar or the weather.

Still, if you are in the Northern Hemisphere, you can also determine which season it is from the shadow line or **terminator** on the First Quarter Moon.

On or around an equinox day, with the Sun shining directly over the Equator, the **profile of the Moon** at First Quarter Moon, is the exact same illumination as that of the Earth, with the terminator of the Moon straight up and down.

At Summer Solstice, the Sun will be shining directly over the [Tropic of Cancer](#), and the shadow line on the First Quarter Moon will be “tilted back”, as if the Moon were smiling.

Second Extended Study: Our Solar System is an undistinguished star-planetary system in the **Milky Way Galaxy**.



The [Milky Way](#) is a barred spiral [galaxy](#) (or “star city”) much like [Messier 83](#), the **Southern Pinwheel Galaxy** pictured to the left.

Up until the 1920s, [many astronomers](#) believed that the entire Universe consisted of the Milky Way Galaxy and a few smudgy gas cloud formations known as **nebulae** (because of their nebulous or uncertain nature).

In the 1920s, the work of [Edwin Hubble](#) and others showed that the Universe was clearly vastly larger, and that most of the so-called nebulae were in fact completely separate, massive, and relatively far away galaxies.

We now know that the Milky Way is part of the [Local Group](#) of galaxies. The **Local Group** consists of **over 50 galaxies** of various sizes and types.



For example, the **Large Magellanic Cloud** ([LMC](#)) and **Small Magellanic Cloud** ([SMC](#)) are relatively small irregularly shaped Local Group galaxies that can be easily seen in the Southern Hemisphere. [These galaxies were discovered by [Ferdinand Magellan](#), as he passed through the [Straits of Magellan](#), hence their names.]

In the panoramic picture above, we see the Milky Way “edge on” (because we are in it), with the LMC and SMC respectively the large and small smudges to the left. Astronomers believe that the Milky Way and the LMC will merge in about 2 billion years.

The **Andromeda Galaxy** ([M31](#)) a spiral galaxy, is also one of the Local Group of galaxies. Regarding the possible merger of the Andromeda and Milky Way Galaxies, the current best guess is that there is a 50% possibility of that happening in 6 to 7 billion years.



By contrast, and much farther away, we see the **Supergiant Elliptical Galaxy Virgo A** ([M87](#)).

This galaxy is [elliptical](#), meaning it has no spiral arms to distinguish it (it’s basically a blob of stars held together by gravity).



Recently, the black hole at the center of M87 was successfully imaged (see the image to the left), using an [array of eight radio telescopes](#) from around the world collectively called the **Event Horizon Telescope** (EHT).

The EVT has an angular acuity of 20 micro-arcseconds of angle, enough precision to read a newspaper in New York from a sidewalk cafe in Paris.